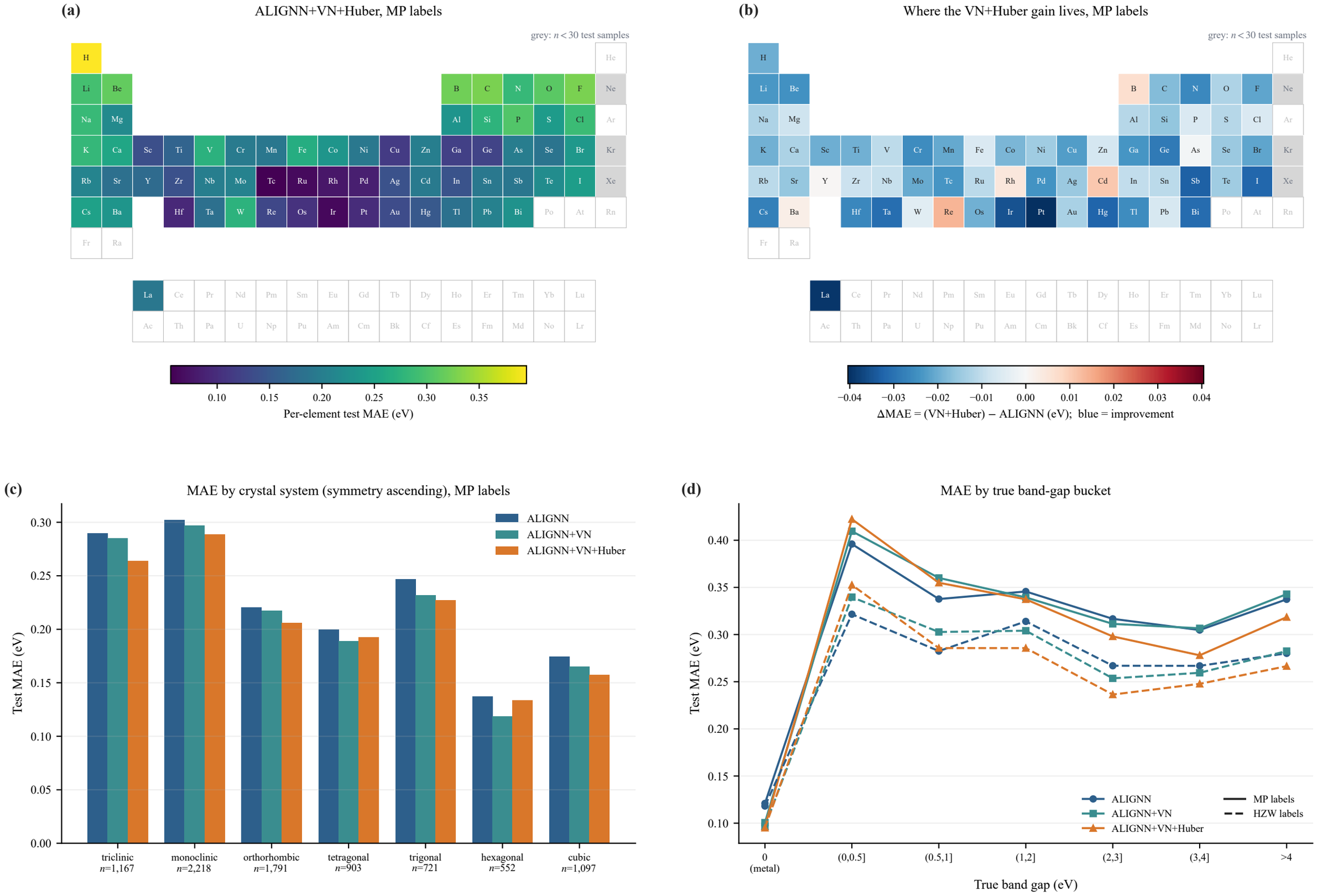

(a)
ALIGNN+VN+Huber, MP labels
grey: n < 30 test samples
0.10
0.15
0.20
0.25
0.30
0.35
Per-element test MAE (eV)
(b)
Where the VN+Huber gain lives, MP labels
grey: n < 30 test samples
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
ΔMAE = (VN+Huber) − ALIGNN (eV); blue = improvement
(c)
MAE by crystal system (symmetry ascending), MP labels
ALIGNN
ALIGNN+VN
ALIGNN+VN+Huber
Test MAE (eV)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
triclinic
n=1,167
monoclinic
n=2,218
orthorhombic
n=1,791
tetragonal
n=903
trigonal
n=721
hexagonal
n=552
cubic
n=1,097
(d)
MAE by true band-gap bucket
Test MAE (eV)
0.10
0.15
0.20
0.25
0.30
0.35
0.40
ALIGNN
ALIGNN+VN
ALIGNN+VN+Huber
MP labels
HZW labels
0
(metal)
(0,0.5]
(0.5,1]
(1,2]
(2,3]
(3,4]
>4
True band gap (eV)